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Nonlinear forward and reconstruction framework for terahertz computed tomography

ZHENGUANG CHEN,[1,3] ZHIGUO WANG,[1,6] BOHUI CHEN,[1,4] AND LIGUO ZHU[2,5]

[1]School of Mathematics, Sichuan University, Chengdu, Sichuan, China
[2]Institute of Fluid Physics, China Academy of Engineering Physics, Mianyang, Sichuan, China
[3]zhenguangchenscu@163.com
[4]chenbohui@scu.edu.cn
[5]zhuliguo@tsinghua.org.cn
[6]wangzg315@126.com

Abstract: Terahertz computed tomography (THz-CT) aims to reconstruct the internal structure of objects using non-ionizing THz waves, offering broad potential for industrial non-destructive testing. However, most existing methods adopt the straight-line propagation assumption as in X-ray CT, neglecting the significant refraction of THz waves in inhomogeneous media, which limits reconstruction accuracy. To address this issue, we propose a nonlinear forward modeling and reconstruction framework. Specifically, we develop an iterative ray propagation algorithm based on Snell's law to model nonlinear THz propagation paths involving both unknown refractive indices and interfaces. Based on this nonlinear forward model, we formulate an inverse problem for reconstructing the refractive index. This problem is highly ill-posed, nonconvex, and nonsmooth, making it challenging for gradient-based methods; therefore, we adopt a derivative-free differential evolution algorithm for robust reconstruction. Once the refractive index and interface geometry are recovered, the nonlinear THz path can be accurately computed. We then formulate the transmission coefficient recovery as a total variation (TV)-regularized optimization problem and solve it efficiently using the alternating direction method of multipliers (ADMM). Experimental results on real THz datasets demonstrate that our method significantly outperforms existing linear-ray-based approaches in reconstruction accuracy.

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1. Introduction

Terahertz (THz) radiation refers to electromagnetic waves in the frequency range of 0.1–10 THz. Owing to its non-ionizing nature and strong penetration capability through many non-metallic and non-polar materials [1], Terahertz radiation has been widely applied in fields such as industrial non-destructive testing and biomedicine [2–6]. To reveal the internal three-dimensional (3D) structure of objects, terahertz computed tomography has been developed [7], which enables volumetric imaging by combining THz sensing with tomographic reconstruction techniques. Generally, THz tomography can operate in either transmission or reflection mode; in this work, we focus on the transmission mode.

Computed tomography (CT) is an imaging technique that reconstructs cross-sectional images of an object by combining a series of transmission measurements acquired from multiple viewing angles. Among conventional modalities, X-ray and ultrasound are widely used for internal structural inspection due to their mature imaging technologies and broad applicability. In contrast, the development of THz-CT has been relatively slow. It was first proposed by Fergusson et al. in 2002 [7], offering new possibilities by enabling the acquisition of both amplitude and spectral phase information—an advantage not available in traditional X-ray imaging.

In THz computed tomography, two major modalities are commonly employed: continuous-wave (CW) [8] and time-domain spectroscopy (TDS) [9]. CW THz systems typically operate at

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fixed frequencies and mainly provide intensity information [10]. Their primary advantages lie in system simplicity, compactness, lower cost, and faster image acquisition, since they do not require time-delay scanning or complex pump–probe optical setups. However, CW imaging lacks depth resolution and does not provide spectral or temporal information, which limits its applicability in certain contexts. In contrast, TDS-based THz imaging first demonstrated by Nagashima and Hangyo in 2001, which utilizes pulsed THz radiation and time-domain measurements to extract a richer set of information [11]. These include depth profiles, absorption spectra, and scattering characteristics of the sample, making TDS a more versatile and informative technique. In this work, we adopt a TDS-based THz CT system to leverage its capability for detailed structural and material characterization.

In THz TDS computed tomography, the measurement data typically take the form of a sinogram, which can be interpreted as a set of projections modeled by the Radon transform [9]. These measurements are intrinsically dependent on the object's internal refractive index and absorption coefficient [9], as the THz wave experiences attenuation and phase delay while propagating through the medium. The goal of THz CT is to reconstruct these spatially varying properties from the observed sinogram, thereby revealing the internal structure of the object.

An accurate forward model of THz wave propagation is essential for reliable image reconstruction. However, most existing THz CT studies adopt the straight-line propagation assumption commonly used in X-ray CT, neglecting the fact that THz waves are subject to significant refraction when encountering regions with varying refractive indices [7]. According to Snell's law, the propagation path of THz radiation bends at material interfaces, resulting in nonlinear trajectories that deviate from straight lines [12]. To more faithfully capture the underlying physics, it is necessary to construct a nonlinear propagation model that accounts for these refractive effects. Based on this improved forward model, we formulate the corresponding inverse problem to recover the internal structure of the object with higher fidelity.

1.1. Related works

CT based on X-ray. In terms of reconstructing cross-sectional images, most existing approaches, including those mentioned in the literature (both continuous wave and pulsed), rely on methods originally developed for X-ray computed tomography. X-ray CT reconstruction research focuses on imaging efficiency [13], multi-spectral CT [14], low-dose imaging [15,16] and limited angle tomography [17], primarily using the Radon transform, which models projections as line integrals of the object's properties along straight paths. As the most mature analytical reconstruction algorithm for X-ray CT, filtered back projection (FBP) is widely adopted due to its computational efficiency and ease of implementation [18]. However, its reliance on complete data and high sensitivity to noise have led researchers to explore more advanced iterative methods. In contrast, the algebraic reconstruction technique (ART) can effectively handle incomplete or noisy data through iterative optimization [19]. When combined with total variation (TV) regularization, its sparsity constraints further suppress noise and preserve edge sharpness, significantly enhancing reconstruction quality in low-dose or undersampled scenarios [20]. In addition, an increasing number of algorithms have been proposed to efficiently solve image reconstruction and restoration problems based on the TV regularization. These include methods such as gradient-flow-based finite element methods [21,22], primal–dual algorithm [23,24], and alternating direction method of multipliers (ADMM) [25–27]. These methods effectively exploit the structural information of the problem while providing convergence guarantees.

CT based on THz ray. In the field of THz CT, incorporating a priori information is essential for enhancing imaging quality. More sophisticated models have been proposed, including solutions to Helmholtz's wave equation [9] and refraction-aware methods based on Snell's law [12]. And in the terahertz band, the generally higher refractive index of the object creates a nonlinear path problem that is difficult to ignore. Tepe et al. proposed a modified algebraic

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reconstruction technique (M-ART) that explicitly accounts for refraction and reflection losses using Snell's law and Fresnel equations, enabling accurate reconstruction of the complex refractive index in terahertz tomography for non-destructive testing of plastics and ceramics [12]. Fosodeder et al. developed a refraction-corrected THz-CT method using ray-tracing and algebraic reconstruction, achieving sub-150μm accuracy in complex plastic profiles [28]. Fosodeder et al. developed a refraction-corrected THz-CT method using a TeraFlash smart THz-TDS system (1600 pulses/s, >3.5 THz bandwidth, 49 fs time resolution) combined with ray-tracing and algebraic reconstruction. This system configuration, featuring a measured beam waist of 0.49 mm, enabled the achievement of sub-150 μm accuracy in complex plastic profiles [28]. Chen et al. indirectly applied such information by introducing a liquid medium with a refractive index matched to that of the sample, thereby mitigating refraction effects when imaging high-refractive-index materials [29]. Paramonov et al. developed a ray-tracing-based simulation method that employs a refractive index scalar field to predict the THz wave propagation path, particularly suitable for media with smoothly varying refractive indices [30]. Building on this approach, May et al. integrated a priori sample modeling with ray-tracing techniques and proposed two reconstruction methods to further improve image quality [31]. Despite their effectiveness, these methods often require strict experimental setups or detailed knowledge of the sample's interface. However, such a priori information about interface are not only difficult to acquire but also presents significant challenges for direct application.

1.2. Our contributions

In this paper, we develop a new nonlinear forward model based on temporal information acquired from THz-TDS. This nonlinear forward model is inspired by [12,28]. Compared with [12,28], our proposed method not only enables the reconstruction of piecewise constant refractive indices but also allows for the simultaneous estimation of partial prior embedded unknown interfaces.

The main contributions of this work are summarized as follows:

- **Nonlinear forward model**. We propose an iterative ray propagation algorithm that accurately captures the nonlinear behavior of THz wave propagation based on Snell's law, accommodating arbitrarily parameterized refractive indices and interfaces. In contrast to the linear propagation assumptions commonly used in X-ray CT, our model accounts for the refractive nature of THz waves. Building on this, we develop a nonlinear forward model based on THz-TDS, which maps the refractive index and interface parameters directly to a sinusoidal function space.

- **Nonconvex and nonsmooth optimization problem for refractive index reconstruction**. When both the refractive index and interface geometry are unknown, the reconstruction problem can be formulated as a nonlinear least-squares problem. However, this inverse problem is severely ill-posed, highly nonconvex, and nonsmooth, leading to instability in gradient-based optimization. Then we use a derivative-free optimization method, the differential evolution algorithm, to solve the resulting inverse problem. This approach enables the simultaneous recovery of both the refractive index and interface parameters.

- **Transmission coefficient reconstruction via TV-Regularized optimization**. Once the refractive index and interface geometry have been obtained, the nonlinear forward model can be used to accurately compute the coefficient matrix that encodes the path lengths of the rays. Based on this, we formulate a transmission coefficient reconstruction problem as a TV–regularized optimization model, which promotes piecewise robustness to noise. The resulting problem is efficiently solved using the ADMM algorithm [25], enabling accurate recovery of the transmission coefficient.

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We experimentally validate the effectiveness of the proposed method on a THz-TDS system for two nested PE circulars data set and 3D-printed resin THZ model data set. Results show that the proposed methods outperform state-of-the-art methods with linear ray.

1.3. Outline of the paper

The paper is organized as follows. Following an introduction to the basic THz imaging model in Section 2, Section 3 considers the refraction-based nonlinear forward problem. We begin by studying the propagation behavior of terahertz waves and provide a brief overview of the associated mathematical model and algorithm. Then we present in Section 4 the corresponding algorithm for solving the associated inverse problem. Finally, in Section 5, experimental results demonstrate that our time-delayed nonlinear algorithm achieves high-quality imaging for parameterizable models. Moreover, the results verify that absorption coefficient imaging, after matrix-based correction of the optical path system, significantly enhances reconstruction quality.

2. Mathematical modeling of THz tomography

THz-TDS enables the reconstruction of an object's internal structure by its complex refractive index. Let $\Omega \subset \mathbb{R}^n$ be the region of interest that represents the object. Here we assume Ω is an open domain of $\mathbb{R}^n$. The complex refractive index is denoted as a complex function on the region of interest $\tilde{n} : \Omega \to \mathbb{C}$ given by

$$\tilde{n}(x) = n(\mathbf{x}) + i\kappa(x). \tag{1}$$

The real part $n(\mathbf{x})$ corresponds to the ratio

$$n(\mathbf{x}) = \frac{c_0}{c(\mathbf{x})}, \tag{2}$$

where c_0 is the speed of light in a vacuum and c denotes the propagation velocity of the THz wave within the material. The imaginary part $\kappa(\mathbf{x})$ quantifies the absorption of the wave and is related to the absorption coefficient $\alpha(\mathbf{x})$ as

$$\kappa(\mathbf{x}) = \alpha(\mathbf{x})\frac{c_0}{4\pi\nu}, \tag{3}$$

where ν is the frequency of the THz wave. We may treat $\tilde{n}$ as a function on $\mathbb{R}^n$ by assuming $\tilde{n}(\mathbf{x}) = 1$ outside Ω.

In THz-TDS, two types of measured data can be obtained: the optical path difference $\Delta d = c_0(T - T_0)$ and the transmission coefficient $\tau = I/I_0$. Here, T and T_0 represent the travel time with and without the object in the ray path respectively, I_0 denotes the initial intensity and I is the intensity measured at the detector.

We now give the expressions of Δd and τ in terms of $\tilde{n}$. For the local propagation process within the same material which is with constant $c = c_M$, we have

$$T = \frac{|L|}{c_M} \quad \text{and} \quad T_0 = \frac{|L_r|}{c_0},$$

where $|L|$ denotes the distance the THz beam propagates through the material and $|L_r|$ represents the straight-line reference distance in the absence of the material. Then

$$\Delta d = c_0\left(\frac{|L|}{c_M} - \frac{|L_r|}{c_0}\right) = n_M|L| - |L_r|, \tag{4}$$

where $n_M = c_0/c_M$ is the constant refractive index of the material. For the general case, we have

$$\mathrm{g}_{\mathrm{ref}}(L) := \Delta d = \int_L n(\mathbf{x})ds - \int_{L_r} ds. \tag{5}$$

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The transmission coefficient τ, which accounts for absorption, adheres to the principles outlined in Lambert Beer's law:

$$\tau = \frac{I}{I_0} = \exp\left(-\int_L \alpha(\mathbf{x})ds\right). \tag{6}$$

It can be rewritten as

$$\mathrm{g}_{\mathrm{abs}}(L) := \log\frac{1}{\tau} = \int_L \alpha(\mathbf{x})ds. \tag{7}$$

The quantities $\mathrm{g}_{\mathrm{ref}}$ and $\mathrm{g}_{\mathrm{abs}}$, derived from the measured time-domain signals, serve as the fundamental inputs to our two-stage reconstruction framework and possess distinct physical interpretations. $\mathrm{g}_{\mathrm{ref}}$ represents the path difference, which quantifies the accumulated path difference of the THz pulse when traveling through the sample compared to the air reference, making it the core measurement used to reconstruct the refractive index n image. Furthermore, $\mathrm{g}_{\mathrm{abs}}$, derived from the Beer–Lambert law, represents the logarithmic attenuation and physically corresponds to the line integral of the absorption coefficient α along the path L. This quantity is essential for the reconstruction of the absorption coefficient image α.

3. Nonlinear forward problem based on refraction and interfaces

THz wave propagation is determined by both refraction and the geometry of interfaces between different media. We assume interfaces separate the interested domain into several connected components and interfaces become the walls between components. Suppose *the refractive indices are constants within the connected components.* The propagation paths then are continuous and composed of straight-line segments refracted at the various interfaces. In this section, we first formulate the forward model by explicitly characterizing the nonlinear path L geometry induced by refraction and interface interactions. We then derive the exact measurement models based on time delay (5) and transmission coefficient (7), respectively. In this paper, we only consider the case that Ω as a radius R disc in $\mathbb{R}^2$, i.e, $\Omega = \{(x, y)|x^2 + y^2 \leq R^2\}$.

3.1. THz propagation path

Before discussing the terahertz propagation problem, we first provide the necessary symbolic explanations, beginning with the information of the interface within the object being measured. Assume there are Ξ interfaces, denoted as $\tilde{\mathcal{S}} = \{\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_\Xi\}$. Each interface $\mathcal{S}_\xi$, $\xi = 1, \ldots, \Xi$, can be represented by parameterized equations of lines or curves. For example, the ξ-th straight line segment can be mathematically expressed as: $\mathcal{S}_\xi = \left\{(x, y) \mid y = k_\xi x + d_\xi,\ (x, y) \in \Omega,\ x \in [x_\xi^{\min}, x_\xi^{\max}]\right\}$, where k_ξ and d_ξ denote the slope and intercept parameters of this line, and $[x_\xi^{\min}, x_\xi^{\max}]$ defines the finite extent of the segment within the region of interest Ω. Based on this interface structure, we construct the nonlinear propagation path, which consists of piecewise linear segments induced by refraction at each interface.

For the initial propagation segment L_0, starting from point $p_0 = (x_0, y_0)$ with directional vector $w_0 = (a_0, b_0)$, the ray path can be mathematically expressed as:

$$L_0 = \left\{(x, y)|(x, y) = p_0 + tw_0, t \in [0, t_{\xi_0}]\right\}, \tag{8}$$

where t_{ξ_0} represents the **path length** of the segment L_0 until the ray encounters the first material interface. Note that this parameter is proportional to the propagation time via the phase velocity of the wave in the medium. And ξ_0 represents the index of the first interface intersected by the path L_0, i.e. $\xi_0 \in \{1, 2, \ldots, \Xi\}$. Subsequently, for each successive segment L_v ($v = 1, \ldots, V$), the

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ray path is defined as:

$$L_v = \left\{(x, y)|(x, y) = p_v + tw_v, t \in [0, t_{\xi_v}]\right\}, \tag{9}$$

where p_v denotes the starting point and w_v represents the direction vector of the v-th segment. The parameter t_{ξ_v} corresponds to the first intersection point between the ray path and the nearest material interface curve $\mathcal{S}_{\xi_v}$ from the set of interfaces $\tilde{\mathcal{S}}$.

At the interface $\mathcal{S}_{\xi_v}$ between two media with refractive indices n_v and n_{v+1}, the ray propagation transitions from segment L_v to L_{v+1} as illustrated in Fig. 1. The angle of incidence $\theta_{1,v}$ at point p_{v+1} can be determined from the dot product of the incident ray direction w_v and the interface normal $\mathfrak{n}_{p_{v+1}}$:

$$\theta_{1,v} = \arccos \frac{|\langle w_v, \mathfrak{n}_{p_{v+1}} \rangle|}{|w_v||\mathfrak{n}_{p_{v+1}}|}. \tag{10}$$

According to Snell's law, the angle of refraction $\theta_{2,v}$ is given by:

$$\theta_{2,v} = \arcsin\left(\frac{n_v \sin\theta_{1,v}}{n_{v+1}}\right). \tag{11}$$

Then, the angular deviation $\Delta\theta_v$ between the incident and refracted rays can be expressed as:

$$\Delta\theta_v = \pm(\theta_{1,v} - \theta_{2,v}) = \pm\left(\theta_{1,v} - \arcsin\left(\frac{n_v \sin\theta_{1,v}}{n_{v+1}}\right)\right), \tag{12}$$

where the sign depends on the relative orientation of the interface normal and incident ray. The recursive relationship between successive ray segments can be formalized in the following proposition.

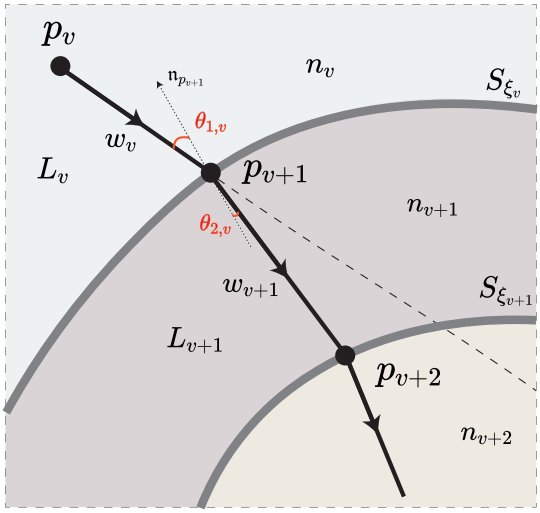


Fig. 1. A schematic of the propagation calculation for a hierarchical structure is shown. The thick grey curve indicates the interface $\mathcal{S}_{\xi_v}$ and $\mathcal{S}_{\xi_{v+1}}$.

For an interface $\mathcal{S}_{\xi_v}$ separating media with refractive indices n_v and n_{v+1}. Based on the v-th segment L_v and the angular deviation $R_{\Delta\theta_v}$, the $(v+1)$-th ray segment $L_{v+1} = \left\{(x, y)|(x, y) = p_{v+1} + tw_{v+1}, t \in [0, t_{\xi_{v+1}}]\right\}$ can be determined by:

$$\begin{aligned} p_{v+1} &= p_v + t_{\xi_v} w_v, \\ w_{v+1} &= R_{\Delta\theta_v} w_v, \end{aligned} \tag{13}$$

where p_{v+1} is the intersection point with the interface, w_{v+1} is the new direction vector, and

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Algorithm 1. Iterative Ray Propagation Algorithm: $L_v \to L_{v+1}$

Require:

- Current ray segment: $L_v(t) = \{(x, y)|(x, y) = p_v + tw_v, t \in [0, t_{\xi_v}]\}$.
- Interface set: $\tilde{\mathcal{S}}$.
- Local refractive indices: n_v, n_{v+1}.

Ensure: Next ray segment L_{v+1}.

1: Compute intersection point: $p_{v+1} = p_v + t_{\xi_v} w_v$,
2: Calculate incident angle: $\theta_{1,v} = \arccos \frac{|\langle w_v, \mathrm{n}_{p_{v+1}} \rangle|}{|w_v||\mathrm{n}_{p_{v+1}}|}$,
3: Apply Snell's law: $\theta_{2,v} = \arcsin\left(\frac{n_v \sin\theta_{1,v}}{n_{v+1}}\right)$,
4: Compute angular deviation: $\Delta\theta_v = \pm(\theta_{1,v} - \theta_{2,v})$,
5: Update direction vector: $w_{v+1} = R_{\Delta\theta_v} w_v$,
6: Define next segment: $L_{v+1}(t) = \{(x, y)|(x, y) = p_{v+1} + tw_{v+1}, t \in [0, t_{\xi_{v+1}}]\}$.

$$R_{\Delta\theta_v} = \begin{bmatrix} \cos\Delta\theta_v & -\sin\Delta\theta_v \\ \sin\Delta\theta_v & \cos\Delta\theta_v \end{bmatrix}, \tag{14}$$

is the rotation matrix corresponding to the angular deviation $\Delta\theta_v$ defined in (12). The pseudo-code for computing the iterative propagation L_{v+1} satisfying Snell's law is as in Algorithm 1.

3.2. Forward problem of time delay

Based on the measurement models established in Section 2, the time delay path difference measurement $\mathrm{g}_{\mathrm{ref}}(L)$ for complex objects can be determined through Eq. (5). The critical challenge lies in evaluating the integral $\int_L n(x)\,dx$, where the ray path L itself is dependent on both the refractive index distribution $n(x)$ and the interface geometry $\tilde{\mathcal{S}}$. While a closed-form solution for this integral is generally intractable due to the complex ray path dependencies, we can derive an analytical expression under certain conditions. Let the refractive index $n(x, y)$ be a piecewise constant function taking values in the set $\{n_0, n_1, \ldots, n_V\}$, where $n_0 = 1$ represents the refractive index of air, and let the initial propagation vector w_0 be normalized such that $|w_0| = 1$. Then the time delay measurement can be expressed as:

$$\mathrm{g}_{\mathrm{ref}} = \sum_{v=0}^{V} t_{\xi_v} n_v - |L_r|, \tag{15}$$

where n_v is the refractive index corresponding to segment L_v, V denotes the total number of interface intersections along the ray path, $|L_r|$ is the length of the reference path in air, and t_{ξ_v} represents the path length of segment L_v, determined by the interface set $\tilde{\mathcal{S}}$ and refractive index distribution $n(x, y)$. One can observe that the time delay measurement is the function of both the refractive index and the path length t_{ξ_v}. Then we can define a nonlinear forward mapping:

$$\mathcal{H}(\mathrm{N}, \mathrm{S}) := \sum_{v=0}^{V} t_{\xi_v} n_v - |L_r|, \tag{16}$$

where $\mathcal{H}$ represents the nonlinear forward operator, $\mathrm{N} = (n_1, n_2, \ldots, n_V)$ is the vector of refractive indices, $\mathrm{S} = (\mathrm{S}_1, \mathrm{S}_2, \ldots, \mathrm{S}_\Xi)$ denotes the interface parameterization, and Ξ represents the total number of interfaces in the reconstruction domain. In (16), the nonlinear forward operator $\mathcal{H}$

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is explicitly defined as a function of the object parameters (N, S) and the ray initial conditions (p_0, w_0). The path lengths $\{t_{\xi_\nu}\}$ are determined by the ray-tracing procedure (Algorithm 1) for a given set of these parameters.

Remark 3.1 *In our forward model, we explicitly account for the nonlinearity in THz wave propagation caused by refraction. To this end, we parameterize the primary factors influencing the light path: the refractive index and the interface geometry. Unlike models that assume a known interface [12,28], our approach retains partial a priori structural information while enabling a flexible, parameterized representation of the interface.*

3.3. Forward problem of transmission coefficient

Based on Eq. (7), the transmission measurement data $g_{abs}(L)$ is related to the absorption coefficient $\alpha(x, y)$ through:

$$g_{abs}(L) = \log \frac{1}{\tau} = \int_L \alpha(x, y) ds. \tag{17}$$

A key observation is that the ray path L is uniquely determined by the refractive index distribution $n(x, y)$ and the material interface geometry $\tilde{\mathcal{S}}$. This implies that L is independent of the absorption coefficient $\alpha(x, y)$. This decoupling allows us to treat the path L as known once the refractive properties are determined, similar to the approach used in X-ray CT. However, a significant difference is that THz-CT utilizes a nonlinear path L rather than a linear path. As a result, we can linearize this integral by discretizing the absorption coefficient distribution.

To this end, we superimpose a square grid over the domain of $\alpha(x, y)$ and assume that the absorption coefficient is piecewise constant within each grid cell. Let α_j denote the constant value in the j-th cell, and let M be the total number of cells. The THz ray, which follows a nonlinear trajectory through the (x, y)-plane, intersects multiple grid cells. For illustration, the path of the i-th ray in Fig. 2(left).

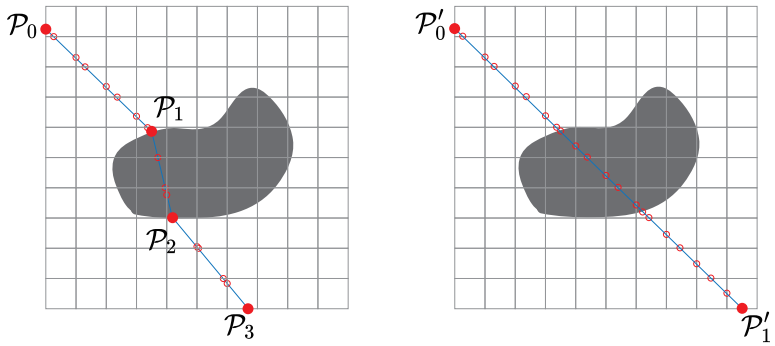


Fig. 2. The left side represents a schematic of terahertz propagation and the right side represents a schematic of X-ray propagation.

The corresponding line integral is referred to as a ray-sum. Denote by g^i_{abs} be the ray-sum measured along the i-th ray. Its relationship with the discretized absorption coefficients is given by:

$$g^i_{abs} = \sum_{j=1}^{J} l_{ij}\alpha_j, \; i \in \{1, 2, \ldots, I\}, \; j \in \{1, 2, \ldots, J\}, \tag{18}$$

where N is the total number of rays and l_{ij} is the weighting factor that represents the contribution of the j-th cell to the i-th ray integral, which is equal to the length of the segment that represents the i-th ray that traverses the j-th cell, J denotes the total number of pixels in the image. Notably, most of the l_{ij} are zero, as each ray intersects only a small subset of the grid cells.

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Equation (18) can be expressed compactly in matrix form as:

$$\mathbf{g}_{\text{abs}} = \mathbf{L}\boldsymbol{\alpha}, \tag{19}$$

where $\mathbf{L}$ is the coefficient matrix containing the path lengths, $\boldsymbol{\alpha}$ is the vector of unknown absorption coefficients, and $\mathbf{g}_{\text{abs}}$ is the vector of transmission measurements. Comparing Fig. 2(left) with Fig. 2(right), we observe that the nonlinear propagation of THz rays results in a different coefficient matrix $\mathbf{L}$ compared to that of X-ray CT, highlighting the distinct characteristics of THz-based imaging.

4. Inverse problems and optimization algorithms

In this section, we first employ the differential evolution algorithm to address the inverse problem of time delay, which enables the estimation of both the refractive index distribution and the material interface geometry. Once these refractive properties are recovered, the corresponding ray path L can be accurately determined. Subsequently, in Subsection 4.2, We use the ADMM algorithm to solve the inverse problem for the absorption coefficient combined with TV regularization.

4.1. Differential evolution algorithm for the inverse problem of refractive index and interface

Given a specific parameterization of the interface, the forward model in (16) describes how the i-th time-delay measurement $\mathrm{g}^i_{\text{ref}}$ is nonlinearly determined by the refractive index distribution N and the interface geometry S through the mapping $\mathcal{H}_i(\mathrm{N}, \mathrm{S})$. Based on this model, the inverse problem aims to estimate the optimal refractive index N^* and interface S^* by solving the following optimization problem:

$$(\mathrm{N}^*, \mathrm{S}^*) = \arg\min_{(\mathrm{N},\mathrm{S})} \sum_{i=1}^{I} \|\mathrm{g}^i_{\text{ref}} - \mathcal{H}_i(\mathrm{N}, \mathrm{S})\|^2, \tag{20}$$

where both N and S are treated as optimization variables.

It is clear that problem (20) is nonconvex and nonsmooth, which makes finding the global optimum highly challenging. Even in the simplest case where the interface is assumed to be linear, the optimization variable reduces to $\mathrm{S} = (k_{\xi_v}, d_{\xi_v})$. However, it is evident that the mapping $\mathcal{H}_i(\mathrm{N}, \mathrm{S})$ is a nonsmooth function of k_{ξ_v}, primarily due to its appearance in the denominator.

To tackle the challenging nonconvex and nonsmooth inverse problem in Eq. (20), we employ the Differential Evolution (DE) algorithm [32]. DE is a population-based stochastic optimization method specifically designed for global optimization over continuous spaces. Its suitability for our problem stems from three key attributes: (1) it is derivative-free, which is essential given the non-differentiable nature of the forward operator $\mathcal{H}_i(\mathbf{N}, \mathbf{S})$ with respect to the interface parameters; (2) it is a global optimizer, which reduces the risk of converging to poor local minima, a significant concern in nonconvex problems; and (3) it does not require a single, precise initial guess. Instead, DE initializes a population of candidate solutions by sampling uniformly within a predefined parameter search domain. This domain is defined by setting physically plausible lower and upper bounds for each parameter in $(\mathbf{N}, \mathbf{S})$, such as the expected range of refractive indices for the material and the spatial extent of the reconstruction domain for interface locations. The algorithm then iteratively improves the population through cycles of mutation, crossover, and selection.

The key hyperparameters for DE are the population size (NP), the crossover probability (CR), and the mutation scaling factor (λ). In our experiments, we set $\mathrm{NP} = 10 \times D$, where D is the number of parameters in $(\mathbf{N}, \mathbf{S})$, $CR = 0.9$, and $\lambda = 0.8$. These values are chosen based on common practices in the literature and were found to provide a good balance between exploration

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and convergence speed for our problems. The detailed steps of the DE algorithm are outlined as follows.

Step 1: Initialization. Randomly generate an initial population of NP candidate solutions:

$$\{(\mathrm{N}_m^0, \mathrm{S}_m^0)\}_{m=1}^{\mathrm{NP}},$$

where $(\mathrm{N}_m^0, \mathrm{S}_m^0)$ are sampled uniformly within the prescribed feasible region. This initial population provides a diverse set of starting points for the search process, which is essential for exploring the solution space thoroughly.

Step 2: Mutation. For each target vector $(\mathrm{N}_m^k, \mathrm{S}_m^k)$ at iteration k, generate a mutant vector by:

$$(\mathrm{N}_m^{k,\mathrm{mut}}, \mathrm{S}_m^{k,\mathrm{mut}}) = (\mathrm{N}_{r_1}^k, \mathrm{S}_{r_1}^k) + \lambda \cdot \left[(\mathrm{N}_{r_2}^k, \mathrm{S}_{r_2}^k) - (\mathrm{N}_{r_3}^k, \mathrm{S}_{r_3}^k)\right],$$

where r_1, r_2, r_3 are distinct indices randomly selected from $\{1, 2, \ldots, \mathrm{NP}\}$, and λ>0 is the mutation scaling factor. This step introduces diversity into the population, helping the algorithm explore different regions of the solution space.

Step 3: Crossover. Generate a trial vector by mixing the target vector and the mutant vector:

$$(\mathrm{N}_m^{k,\mathrm{trial}}, \mathrm{S}_m^{k,\mathrm{trial}}) = \begin{cases} (\mathrm{N}_m^{k,\mathrm{mut}}, \mathrm{S}_m^{k,\mathrm{mut}}), & \text{if Rand} \le \mathrm{CR} \text{ or } j = j_{\mathrm{rand}}, \\ (\mathrm{N}_m^k, \mathrm{S}_m^k), & \text{otherwise}, \end{cases}$$

where $\mathrm{CR} \in (0, 1)$ denotes the crossover probability, Rand is a uniformly distributed random number in $[0, 1]$, and j_{rand} is a randomly chosen index to ensure at least one component from the mutant vector is included. This step maintains a balance between exploration and exploitation.

Step 4: Selection. Evaluate the objective function for both the target vector and the trial vector. Update the solution according to the greedy selection rule:

$$(\mathrm{N}_m^{k+1}, \mathrm{S}_m^{k+1}) = \begin{cases} (\mathrm{N}_m^{k,\mathrm{trial}}, \mathrm{S}_m^{k,\mathrm{trial}}), & \text{if } f(\mathrm{N}_m^{k,\mathrm{trial}}, \mathrm{S}_m^{k,\mathrm{trial}}) \le f(\mathrm{N}_m^k, \mathrm{S}_m^k), \\ (\mathrm{N}_m^k, \mathrm{S}_m^k), & \text{otherwise.} \end{cases}$$

This greedy selection process ensures that the population evolves toward better solutions over time.

Step 5: Termination. The algorithm continues to iterate until a predefined maximum number of generations, $K_{\max}$, is reached. The algorithm then outputs the best solution $(\mathrm{N}^*, \mathrm{S}^*)$ found during the search process.

Here, we present Algorithm 2 to illustrate the detailed procedure for solving the time-delay inverse problem.

4.2. ADMM algorithm for the inverse problem of transmission coefficient

After obtaining the refractive index and the interface geometry, we can compute the exact path L. Based on the linear Eq. (3.3), we then formulate the corresponding optimization problem to obtain the transmission coefficient

$$\boldsymbol{\alpha}^* = \arg\min_{\boldsymbol{\alpha}} \left\{ \|\mathbf{L}\boldsymbol{\alpha} - \mathbf{g}_{\mathrm{abs}}\|_2^2 \right\}. \tag{21}$$

However, this is a poorly conditioned inverse problem because I (the number of measurements) is less than J (the number of unknowns variables), i.e., I<J. To address this ill-posedness and stabilize the solution, we introduce a Total Variation (TV) regularization term. Specifically, we consider the following TV-regularized optimization problem:

$$\boldsymbol{\alpha}^* = \arg\min_{\boldsymbol{\alpha}} \left\{ \lambda \|\boldsymbol{\alpha}\|_{\mathrm{TV}} + \|\mathbf{L}\boldsymbol{\alpha} - \mathbf{g}_{\mathrm{abs}}\|_2^2 \right\}, \tag{22}$$

where λ denotes the regularization parameter, which balances the fidelity term with the regularity term as weights. This ill-posed inverse problem can be effectively solved by ADMM algorithm

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Algorithm 2. Solving time-delay inverse problem (20) using DE

Require:

- Measurement data: $\mathbf{g}_{\text{ref}}$.
- Parameter bounds: $\mathcal{B} = \{(\mathbf{N}_{\min}, \mathbf{N}_{\max}), (\mathbf{S}_{\min}, \mathbf{S}_{\max})\}$.

Initialize population $\{(\mathsf{N}_m^0, \mathsf{S}_m^0)\}_{m=1}^M$ in First-generation population $\mathbf{GP}^0$,
while not converged **do**
  for each candidate $(\mathsf{N}^k, \mathsf{S}^k) \in \mathbf{GP}^k$ **do**
    Compute path length t_{ξ_v} of every segment L_v by Algorithm 1,
    Compute $\mathcal{H}_i(\mathsf{N}^k, \mathsf{S}^k)$ using (16),
    Evaluate loss $\mathcal{L}_{(\mathsf{N},\mathsf{S})} = \frac{1}{2}\sum_i^I \|\mathbf{g}_{\text{ref}}^i - \mathcal{H}_i(\mathsf{N}^k, \mathsf{S}^k)\|^2$,
  end for
  Update population $\mathbf{GP}^{k+1}$ via DE algorithm,
end while
return refractive index and interface geometry $(\mathsf{N}^*, \mathsf{S}^*)$.

[25], which overcoming the non-triviality of TV through variable partitioning of the regularization term and the data fidelity term, achieving a closed form in which the regularization parameter is updated in each iteration.

Specifically, auxiliary variables $\boldsymbol{\gamma} = \nabla\boldsymbol{\alpha}$ are introduced in order to separate the variables. Then we can write the equivalent constrained form of problem (22) as follows:

$$\begin{aligned} \boldsymbol{\alpha}^*, \boldsymbol{\gamma}^* = &\arg\min_{\boldsymbol{\alpha},\boldsymbol{\gamma}} \left\{\lambda\|\boldsymbol{\gamma}\|_1 + \|\mathbf{L}\boldsymbol{\alpha} - \mathbf{g}_{\text{abs}}\|_2^2\right\}, \\ &\text{s.t.} \qquad \boldsymbol{\gamma} = \nabla\boldsymbol{\alpha}. \end{aligned} \tag{23}$$

To solve the constrained problem (23), we construct its Augmented Lagrangian(AL). Starting from the standard form with Lagrange multiplier $\boldsymbol{\mu}$:

$$\mathcal{L}_0(\boldsymbol{\alpha}, \boldsymbol{\gamma}, \boldsymbol{\mu}) = \lambda\|\boldsymbol{\gamma}\|_1 + \|\mathbf{L}\boldsymbol{\alpha} - \mathbf{g}_{\text{abs}}\|_2^2 + \boldsymbol{\mu}^\top(\boldsymbol{\gamma} - \nabla\boldsymbol{\alpha}) + \frac{\rho}{2}\|\boldsymbol{\gamma} - \nabla\boldsymbol{\alpha}\|_2^2,$$

we introduce the scaled dual variable $\boldsymbol{u} = \boldsymbol{\mu}/\rho$ and rewrite it as:

$$\mathcal{L}(\boldsymbol{\alpha}, \boldsymbol{\gamma}, \boldsymbol{u}) = \lambda\|\boldsymbol{\gamma}\|_1 + \|\mathbf{L}\boldsymbol{\alpha} - \mathbf{g}_{\text{abs}}\|_2^2 + \frac{\rho}{2}\|\boldsymbol{\gamma} - \nabla\boldsymbol{\alpha} + \boldsymbol{u}\|_2^2 - \frac{\rho}{2}\|\boldsymbol{u}\|_2^2, \tag{24}$$

where ρ>0 is the penalty parameter.

ADMM effectively handles the non-differentiability of the TV norm by decomposing the original problem into subproblems that are easier to solve. To solve the ill-posed inverse problem in (23), we adopt an ADMM-based algorithm that decouples the data fidelity and total variation regularization via variable splitting. The general procedure involves minimizing the AL (24) with respect to each variable while keeping the others fixed. The optimization is carried out iteratively through the following steps. At each iteration k, the variables $\boldsymbol{\alpha}, \boldsymbol{\gamma}, \boldsymbol{u}$ are updated as follows:

Update of $\boldsymbol{\alpha}$: We minimize the $\mathcal{L}$ with respect to $\boldsymbol{\alpha}$, holding $\boldsymbol{\gamma}^k$ and $\boldsymbol{u}^k$ fixed:

$$\boldsymbol{\alpha}^{k+1} = \arg\min_{\boldsymbol{\alpha}} \left\{\|\mathbf{L}\boldsymbol{\alpha} - \mathbf{g}_{\text{abs}}\|_2^2 + \frac{\rho}{2}\|\boldsymbol{\gamma}^k - \nabla\boldsymbol{\alpha} + \boldsymbol{u}^k\|_2^2\right\}. \tag{25}$$

This subproblem is approximately solved by an iterative weighted strategy that balances the contributions of the data fidelity and regularization terms [25,33,34].

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Update of $\boldsymbol{\gamma}$: Next, we minimize $\mathcal{L}$ with respect to $\boldsymbol{\gamma}$, with $\boldsymbol{\alpha}^{k+1}$ and $\boldsymbol{u}^k$ fixed:

$$\boldsymbol{\gamma}^{k+1} = \arg\min_{\boldsymbol{\gamma}} \left\{ \lambda \|\boldsymbol{\gamma}\|_1 + \frac{\rho}{2} \|\boldsymbol{\gamma} - \nabla\boldsymbol{\alpha}^{k+1} + \boldsymbol{u}^k\|_2^2 \right\}. \tag{26}$$

This subproblem corresponds to a proximal mapping of the ℓ_1-norm and admits a closed-form solution via soft-thresholding:

$$\boldsymbol{\gamma}^{k+1} = \mathcal{T}(\nabla\boldsymbol{\alpha}^{k+1} - \boldsymbol{u}^k, \tfrac{\lambda}{\rho}), \tag{27}$$

where the soft-thresholding operator $\mathcal{T}(\cdot, \tau)$ is defined element-wise as:

$$\mathcal{T}(a, \tau) = \text{sign}(a) \cdot \max(|a| - \tau, 0). \tag{28}$$

Update of the scaled dual variable $\boldsymbol{u}$: Finally, the dual variable is updated as follows:

$$\boldsymbol{u}^{k+1} = \boldsymbol{u}^k + \nabla\boldsymbol{\alpha}^{k+1} - \boldsymbol{\gamma}^{k+1}. \tag{29}$$

Here, building on the previously introduced time-delayed forward problem, the refractive index and interfacial parameters estimated through the time-delayed inverse problem are used to compute the correction coefficient matrix $\mathbf{L}$. This correction enables a more accurate reconstruction of the absorption coefficients. Accordingly, the complete procedure is formally outlined in Algorithm 3.

Algorithm 3. ADMM algorithm with corrected matrices for absorption coefficients

Require:

- Measurement data of Time delay: $\mathbf{g}_{\text{ref}}$.
- Measurement data of Absorption coefficient: $\mathbf{g}_{\text{abs}}$.

1: Calculate refractive index and interface parameters $(\mathsf{N}^*, \mathsf{S}^*)$ using Algorithm 1 and 2.
2: Compute $p_0, ..., p_V$ via Algorithm 1.
3: Compute $\mathbf{L}$ using (19).
4: Initialize absorption coefficient $\boldsymbol{\alpha}^0 = 0$.
5: Get $\boldsymbol{\alpha}^*$ via ADMM algorithm with (25)-(29).
6: **return** absorption coefficient $\boldsymbol{\alpha}^*$.

The overall framework of the proposed reconstruction algorithm is illustrated in Fig. 3. In the first stage, the refractive index distribution and interface geometry are recovered from time-delay measurements using a differential evolution algorithm. The second stage then utilizes these parameters to compute the corrected system matrix and reconstructs the absorption coefficient via TV-regularized optimization.

5. Experiment results

To demonstrate the effectiveness of the proposed nonlinear propagation model, we conduct two experiments using different target samples: a nested polyethylene (PE) ring and a 3D-printed resin "THZ" object, see Fig. 5.

In the first experiment, we reconstruct the refractive index from time-delay measurements using the DE algorithm. In the second experiment, we reconstruct the absorption coefficients of the target object using transmission measurements, relying on the nonlinear forward model and the TV-regularized optimization framework solved by ADMM algorithm.

We compare our method against classical approaches, including FBP [18] and M-ART [12]. It is worth noting that M-ART reconstructs the refractive index with known interface information.

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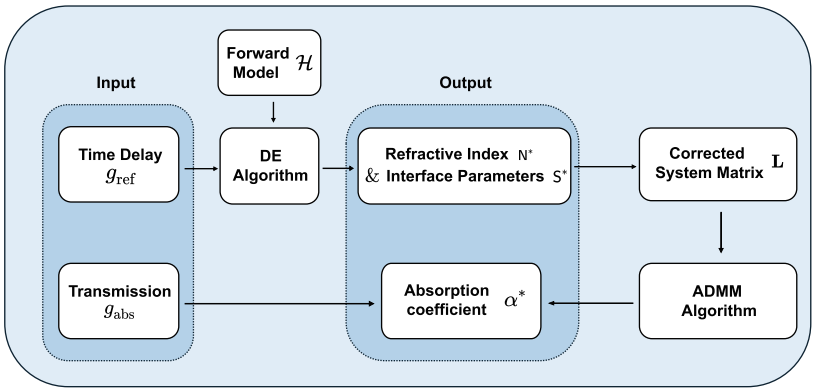


Fig. 3. Schematic diagram of the reconstruction framework for THz computed tomography.

In contrast, our proposed method jointly encodes both the refractive index and parameterized interface information within the model, making it applicable to a broader range of practical scenarios.

Both datasets were acquired using a terahertz time-domain spectrometer synchronized with the scanning platform, as Fig. 5. As the platform rotates θ and shifts s, spectral signals are collected at each specific position. Each measured pulse signal is then compared with a reference signal to extract the time-delay ΔT and transmission coefficient τ information, as Fig. 4. These values are subsequently processed using (5) and (7) to obtain the corresponding measurement g_{ref} and g_{abs}. The results presented below are obtained by running python implementation on personal computer with Intel Core i9-12900K(3.2GHz) and Ram of 64G under Windows 11. We used Peak Signal-to-Noise Ratio (PSNR) [27] and Structural Similarity SSIM [35] to assess the quality of observed and recovered images.

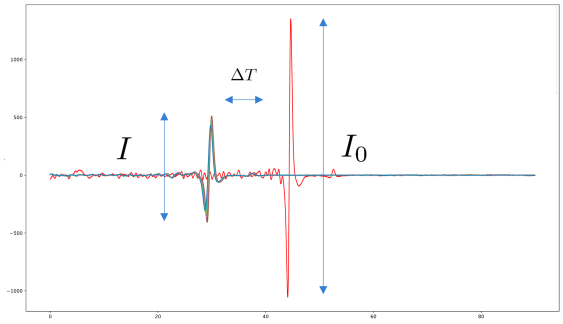
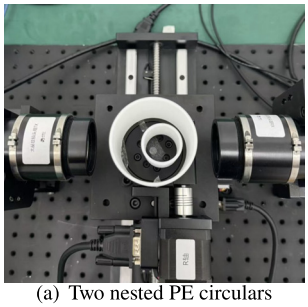


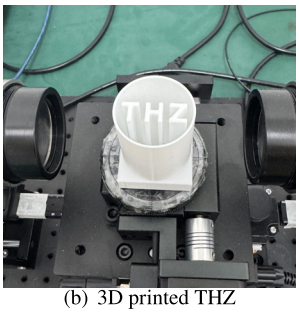
Fig. 4. The time delay ΔT and transmission coefficient $\tau = I/I_0$ are extracted from the pulse signal.

5.1. Two nested PE circulars data set

The purpose of the first experiment is to validate the parameterized forward model. We assume that the object being measured has a piecewise constant refractive index, with both the refractive index and the interface of the reconstructed object parameterized by (N, S). For simplicity, we

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(a) Two nested PE circulars



(b) 3D printed THZ

Fig. 5. The two real objects used for the validation experiments are: (a) a cylindrical shape in PE material with a cross-sectional profile considered as two nested PE circulars; (b) a THZ model 3D printed in resin material.

employ a model comprising two rings, with the parameter set as follows:

$$\begin{aligned} \mathrm{N} &= \{\mathrm{n}_1, \mathrm{n}_2\}, \\ \mathrm{S} &= \mathrm{S}_1 \cup \mathrm{S}_2, \\ \mathrm{S}_\xi &= \{\mathrm{r}_\xi, \mathrm{d}_\xi, \tilde{\mathrm{x}}_\xi, \tilde{\mathrm{y}}_\xi\}, \xi = 1, 2, \end{aligned} \tag{30}$$

where r_ξ represents the inner radius of the ξ-th ring, d_ξ represents the thickness of the ring, and $(\tilde{x}_\xi, \tilde{y}_\xi)$ represents the center coordinates of the ring. Then for the actual measurement of the object, we have

$$(\mathrm{N}, \mathrm{S}) = (\mathrm{n}_1, \mathrm{n}_2, \mathrm{r}_1, \mathrm{d}_1, \tilde{\mathrm{x}}_1, \tilde{\mathrm{y}}_1, \mathrm{r}_2, \mathrm{d}_2, \tilde{\mathrm{x}}_2, \tilde{\mathrm{y}}_2). \tag{31}$$

The optimization problem is then formulated as follows:

$$(\mathrm{N}^*, \mathrm{S}^*) = \arg\min_{(\mathrm{N},\mathrm{S})} \sum_{i=1}^{I} \|\mathrm{g}^i_{\mathrm{ref}} - \mathcal{H}_i(\mathrm{N}, \mathrm{S})\|^2. \tag{32}$$

After data preprocessing, we obtain the pulse waveform of each (s, θ) initialized THz ray after it passes through the object and then extract its time delay information. Following data collection, we extracted measurements at discrete angles of $0°, 10°, 20°, \ldots, 180°$ and s-displacements of $0, 10, 20, \ldots, 70$mm, then $I = 19 \times 8 = 152$. The parameters of the known ground truth model are:

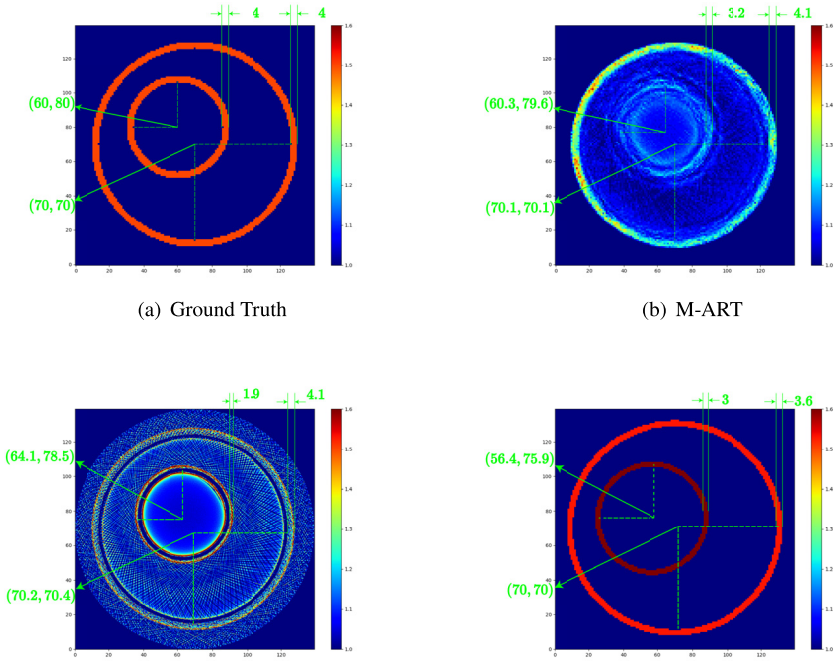
$$(\mathrm{n}_1, \mathrm{n}_2, \mathrm{r}_1, \mathrm{d}_1, \tilde{\mathrm{x}}_1, \tilde{\mathrm{y}}_1, \mathrm{r}_2, \mathrm{d}_2, \tilde{\mathrm{x}}_2, \tilde{\mathrm{y}}_2) = (1.54, 1.54, 26, 4, 60, 80, 60, 4, 70, 70). \tag{33}$$

Note that, except for the refractive index, all other parameter scales are in units of 0.5 mm.

As demonstrated in Fig. 6, the proposed method achieves greater accuracy in parameter estimation compared to traditional approaches. To be specific, FBP and M-ART imaging effectively reconstruct the center coordinates of the two circles when compared to the real image. However, the recovery of thickness and refractive index values remains inadequate. The FBP algorithm, in particular, remains substantial artifacts that degrade imaging quality. The M-ART algorithm significantly improves the reconstruction quality of large circular rings, mainly due to its full integration of prior information about the interface. This strong prior information has rigid requirements for the precise position information of the interface. Compared with the M-ART algorithm, we parameterize the interface and refractive index, applying the DE algorithm to estimate the position information. From Fig. 6(d), we observe that the reconstructed center coordinates of the small circle deviate slightly, but the thickness and refractive index distributions of both circles remain highly accurate compared to the real image. Moreover, our method only relies on the rough shape of the interface to embed the parametric unknown interface into the

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forward model and simultaneously reconstruct the refractive index distributions. This partial prior that only depends on the rough shape of the interface can not only reduce the requirements for the model but also be more applicable to more generalized reconstruction tasks, thereby enhancing the flexibility of the method. In addition, as shown in Table. 1, the proposed method with DE algorithm performs better on both the test metrics PSNR and SSIM.



(a) Ground Truth (b) M-ART

(c) FBP (d) DE

Fig. 6. Comparison of the reconstruction results for two nested polyethylene (PE) circular models using different algorithms, based on real optical path differences.

Table 1. Comparison of the reconstruction effects of different methods for evaluating refractive index fields using PSNR and SSIM.

Method	PSNR	SSIM
FBP	20.23	0.6838
M-ART	20.89	0.8114
Proposed methods	**27.16**	**0.8977**

5.2. THZ data set

In this subsection, we also verified the reconstruction of the object using the extinction coefficient after correcting the light path. To validate the algorithm's efficacy, we materialized the tangible object intended for assessment through 3D printed THZ (see Fig. 5(b)), employing photosensitive resin.

To reconstruct the complex "THZ" structure and handle arbitrary segment orientations, we adopted a direct vertex-based parameterization, defined by the endpoints of each segment.

$$S_\xi = \{A_\xi + t(B_\xi - A_\xi) \mid t \in [0, 1]\},$$

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where the endpoints are denoted by $A_\xi = (x_\xi^A, y_\xi^A)$ and $B_\xi = (x_\xi^B, y_\xi^B)$. For the THz sample, the entire "THZ" structure is decomposed into piecewise linear segments. For example, the letter 'T' represented by 8 key points, 'H' by 12 points, and 'Z' by 10 points. The total parameter vector $\mathrm{S} = \{\mathrm{S}^\mathrm{T}, \mathrm{S}^\mathrm{H}, \mathrm{S}^\mathrm{Z}\}$ for the interfaces is then the concatenation of all these (x, y) coordinates:

$$\begin{aligned}
\mathrm{S}^\mathrm{T} &= [x_1^\mathrm{T}, y_1^\mathrm{T}, x_2^\mathrm{T}, y_2^\mathrm{T}, \ldots, x_8^\mathrm{T}, y_8^\mathrm{T}], \\
\mathrm{S}^\mathrm{H} &= [x_9^\mathrm{H}, y_9^\mathrm{H}, x_{10}^\mathrm{H}, y_{10}^\mathrm{H}, \ldots, x_{20}^\mathrm{H}, y_{20}^\mathrm{H}], \\
\mathrm{S}^\mathrm{Z} &= [x_{21}^\mathrm{Z}, y_{21}^\mathrm{Z}, x_{22}^\mathrm{Z}, y_{22}^\mathrm{Z}, \ldots, x_{30}^\mathrm{Z}, y_{30}^\mathrm{Z}],
\end{aligned}$$

giving a total of 60 parameters to be optimized. We employ the DE algorithm to estimate these parameters. Although DE does not require precise initial guesses for every endpoint, it requires a physically meaningful feasible domain. We define this domain using the spatial bounds of the region of interest (ROI): $x \in [X_{\min}, X_{\max}], y \in [Y_{\min}, Y_{\max}]$.

Although DE is a global optimizer, its efficiency depends on the search-space size. Searching the entire ROI for a high-dimensional parameter vector (as in the THz model) is computationally infeasible. Therefore, we incorporate coarse prior knowledge of the sample's approximate position to restrict each coordinate to a much smaller window. For instance, if the object is roughly centered, each coordinate is constrained within $[X_{\text{center}} - \delta, X_{\text{center}} + \delta]$ and $[Y_{\text{center}} - \delta, Y_{\text{center}} + \delta]$, where $(X_{\text{center}}, Y_{\text{center}})$ denote ground truth of this point, δ is on the order of a few millimeters. This weak prior significantly reduces the effective search space while preserving model flexibility, thereby improving both convergence speed and stability of DE. In our experiments we set $\delta = 5$ mm to balance efficiency and accuracy, and the estimated parameters exhibited errors within 0.5 mm.

The data acquisition process is consistent with that in the previous experiment, where the terahertz time-domain spectrometer is synchronized with the rotating platform to record pulse signals. These signals are subsequently preprocessed to extract the transmission coefficient τ. As shown in Fig. 7, the sinogram of $\mathbf{g}_{\text{abs}}$ is obtained after processing τ using Eq. (7), followed by image reconstruction via the FBP algorithm. The coefficient matrix $\mathbf{L}$ of the THz model is then computed according to Algorithm 1, using the optimized interface parameters obtained from the DE algorithm, and the final image reconstruction is carried out using the ADMM algorithm.

Figure. 8(a) and Fig. 8(b) show the reconstruction results obtained using the FBP and ART algorithms, respectively. The results reveal that conventional algorithms struggle to clearly depict internal interface structures. Both FBP and conventional ART, as imaging methods based on a linear path model, exhibit poor reconstruction quality, capturing only coarse approximations of the letter-shaped features. When we modify the system matrix $\mathbf{L}$ according to a nonlinear ray model, we continues to use ART algorithm to solve problem (21) with a better system matrix $\mathbf{L}$, Fig. 8(c) demonstrates enhanced capability in accurately recovering the absorption distribution within each individual square region. In order to further improve the reconstruction performance, we consider the TV regularization and then solve the problem (22) iteratively using the ADMM algorithm. Figure 8(d) significantly reduces many artifacts and enhances the image quality compared to Fig. 8(c).

Additionally, as shown in Table 2, the proposed method employs a nonlinear propagation model, which corrects the coefficient matrix $\mathbf{L}$ and achieves higher PSNR and SSIM values. This correction significantly enhances image quality. Compared with linear models, the nonlinear formulation provides superior reconstruction performance, while TV regularization further improves structural similarity and suppresses noise by balancing data fidelity and sparsity. The ADMM-based optimization guarantees stable convergence, thereby validating the effectiveness of our approach in terahertz imaging.

While PSNR and SSIM offer a global assessment of imaging quality, they may not fully capture local reconstruction details. To further evaluate local performance, we conduct an error analysis along selected cross-sectional profiles, as shown in Fig. 9. Specifically, relative errors of pixel

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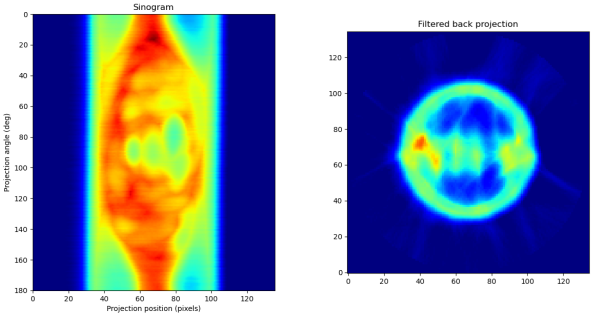
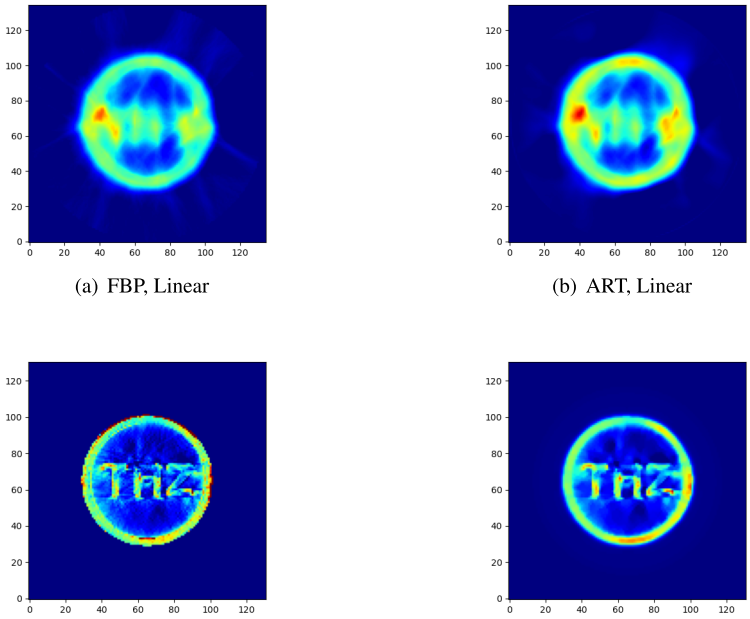


Fig. 7. Transmission coefficient sinograms of 3D printed resin 'THZ' models and the reconstructed results by FBP algorithm.



(a) FBP, Linear (b) ART, Linear

(c) ART, Nonlinear (d) ADMM, Nonlinear

Fig. 8. The transmission coefficient measured from the THZ model printed on photosensitive resin material is compared using the ART and ADMM algorithms with nonlinear forward model.

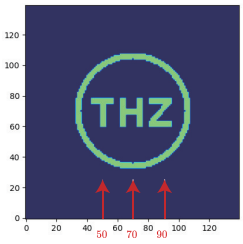
Table 2. Comparison of the reconstruction effects of different methods for evaluating absorption coefficient fields using PSNR and SSIM.

Method	PSNR	SSIM
FBP, Linear	20.34	0.9047
ART, Linear	20.05	0.9020
ART, Nonlinear	23.38	0.9176
ADMM, Nonlinear	**23.47**	**0.9254**

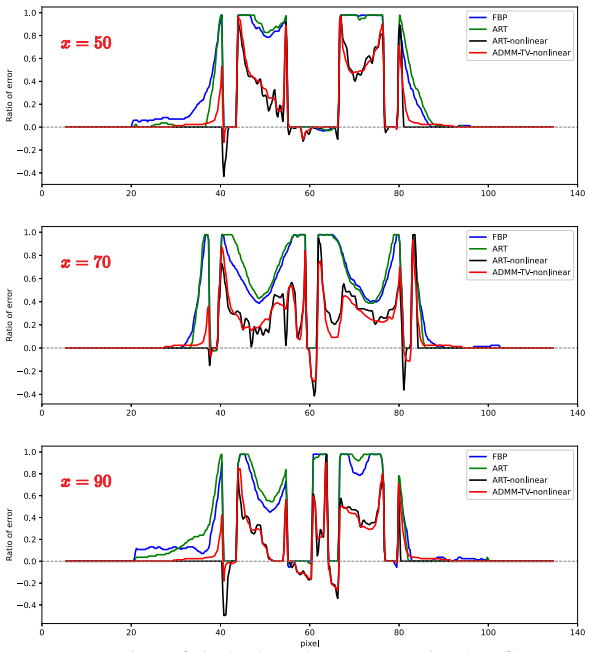
values are computed at $x = 50, 70, 90$ between the reconstructed images and the ground-truth THZ image. The results demonstrate that the proposed nonlinear propagation model consistently

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yields smaller pixel-wise errors compared with the linear model and other baseline methods, highlighting its advantage in reconstruction accuracy and its ability to preserve fine structural details in THZ imaging.



(a) The cross-sectional profiles of the ground-truth THZ image.



(b) Relative error comparison of pixel values along cross-sectional profiles at $x = 50, 70, 90$.

Fig. 9. Comparison of the relative error of the different algorithms. (a) The ground-truth THZ image; (b) The pixel-wise relative error between the reconstructed images obtained by different algorithms and the ground-truth THz image along cross-sectional profiles at $x = 50, 70, 90$.

6. Conclusion

In this work, we developed a nonlinear forward model for THz-TDS that directly maps parameterized refractive indices and interfaces to the sinogram by accurately modeling ray propagation based on Snell's law. Unlike conventional linear CT approaches, our model accommodated both refractive effects and partial prior information, enabling the reconstruction of piecewise constant refractive indices as well as unknown interfaces. To tackle the resulting ill-posed, nonconvex, and nonsmooth inverse problem, we employed a derivative-free differential evolution algorithm, which allows for the simultaneous recovery of refractive indices and interface geometries. Furthermore,

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by leveraging the reconstructed parameters, we formulated a TV-regularized optimization problem for transmission coefficient recovery and solved it efficiently using ADMM. Experimental results on THz-TDS data, including nested PE circulars and a 3D-printed resin model, demonstrated that the proposed approach achieves clearer structures and superior noise suppression compared with state-of-the-art linear-ray methods, thereby validating its effectiveness in realistic scenarios. As the geometric information can be difficult to obtain in some cases, implicit neural representations may offer an interesting future direction.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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